



Ministry of Education

Test Blueprint for Ethiopian Higher Education Institutions’ Exit Examination

Program: - Mechanical Engineering in BSc.

Participant for Blueprint:

- ❖ *Bililign Amare (Addis Ababa University)*
- ❖ *Betelihem Zemedkun (Arba Minch University)*
- ❖ *Dilnesaw Bogale (Arba Minch University)*

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Contents

1. Introduction.....	3
2. Objectives of the blueprint test	3
3. Expected Profiles of Graduates’	4
4. Objectives of the Exit Examination	5
5. Themes, List of Courses and Allocate Credit Hours	6
6. Share of the Themes/Course/Items and KSAs in Percentage	7
7. Learning Outcomes in Terms of Three domain	9
8. Test Blueprint Table	10
9. Conclusion	14

1. Introduction

Mechanical Engineering is a profession that deals with the design, manufacturing, selection, installation, commissioning, operation, and maintenance of all forms of machinery, equipment, and industrial systems. The profession plays a vital role in the establishment and sustainable operation of a nation's manufacturing industries, transport systems, power generation, construction, and mining industries.

To assure those professionals, an exit exam may help a university make pertinent decisions to improve the quality of institution-wide education and, in some cases, can help assess the standard of education in specific courses. Based on the graduation profile, competency, and learning outcomes, some courses are identified and to be included in the exit exam.

Besides the blueprint designed provides a guideline to clearly define the competency of the profession or set of competencies an assessment will measure. It was set by integrating the learning competence of the profession with the learning outcome of each selected courses.

To get a valid test, the theme/course/item share analysis in weight percentage was certainly determined based on the credit hours, knowledge acquired, technical aspects and skills. Moreover, the analytical skill, attitude and managerial skill were also considered as a key skill in the competence of the profession.

2. Objectives of the blueprint test

The blue print designed is significant to identify and to weight the core competencies related to learning outcome. It shall have the following main objectives.

- ✓ To map the core competency with the learning outcome of the selected courses for exit exam.
- ✓ To classify the core competencies based on the knowledge, skills and attitudes.
- ✓ To analyze the weight of themes among each other and also the number of items distributed in each selected course.
- ✓ To specify the number of question items in each learning outcome of the courses.

3. Expected Profiles of Graduates'

Due to his/her strong background, a B.Sc. mechanical engineering graduate can accomplish the following tasks after a brief period of on-job training:

- Represent machine and parts drawing manually and/or with CAD software.
- Understand operating principles of machinery and systems and prepare the specification.
- Design small machinery, piping, and other systems
- Conduct strength analysis of machine components
- Plan production process and assembly of parts
- Determine and optimize production costs
- Determine the layout of machinery and supervise machinery installation.
- Manage the maintenance of equipment
- Control the quality of products.
- Optimize energy utilization in plants.
- Manage the operation of thermal systems, energy conversion and HVAC systems.
- Design, develop, operate, and maintain material handling equipment.

4. Objectives of the Exit Examination

The national public administration exit exam shall have the following objectives

- To produce skilled and competent manpower to national and international market
- Assessing students' educational achievement in major areas of Mechanical Engineering
- Ensuring whether the graduation profile of the Mechanical Engineering curriculum has achieved at least common standards of knowledge and practical skills
- Improving stakeholders' trust and confidence in Mechanical Engineering activities of professionals
- Facilitating the efforts of students to revise the core learning outcomes of the courses covered by the exit examination
- Ensuring all graduates from HEIs satisfy the requirements of the labor market and employability through the national wide implementation of competency-based exit exam
- Creating competitive spirit among Mechanical Engineering departments in Ethiopia with the vies to encouraging them to give due attention to the national standards.

5. Themes, List of Courses and Allocate Credit Hours

Table 1 Themes, List of Courses and Allocate Credit Hours.

Themes	Course	Cr.Hrs.
Engineering Mechanics and Materials	Engineering Materials (I & II)	6
	Strength of Materials (I & II)	6
Engineering Thermo-Fluid System	Fluid Mechanics	3
	Heat Transfer	3
	Refrigeration and Air Conditioning	3
	Engineering Thermodynamics (I & II)	6
Control Engineering	Instrumentation and Measurement	3
	Introduction to Mechatronics	3
Machinery Design and Energy Conversion	IC Engine and Reciprocating Machines	3
	Machine Elements (I & II)	6
	Turbomachinery	3
	Mechanism of Machinery	3
	Materials Handling Equipment	3
Manufacturing, Maintenance and Industrial Management	Manufacturing Engineering (I & II)	6
	Maintenance of Machinery	3
	Industrial Management and Engineering Economy	3

6. Share of the Themes/Course/Items and KSAs in Percentage

Table 2 Learning outcome codes

Learning Outcome Code	Learning outcomes
LO1	Understand the main concepts of engineering materials,
LO2	Explain the influence of crystalline structure on the properties of metal,
LO3	Understand type of defects and explain their influences on the properties of crystals,
LO4	Apply Fick's first and second laws for diffusion and explain application of diffusion in engineering materials
LO5	Explain the causes and main types of plastic deformation, mechanical properties and testing of metals,
LO6	Explain Failure and mechanism of fracture,
LO7	Explain main concepts of phase diagram for different alloys
LO8	Explain phase transformation and kinetics of phase transformation
LO9	Understand basic methods of iron and steel production, types of ferrous metals and effects of alloying elements.
LO10	Explain Heat treatment, analyze applications of heat treatment
LO11	Understand Non-ferrous metals, their engineering application, extraction and processing methods,
LO12	Explain Ceramics, polymer and composite materials, classification, properties and processing methods
LO13	Explain corrosion and degradation in metals/materials
LO14	Analyze the stress-strain relationship in various structural members subjected to single and combined loadings; axial load, torsion, bending and shear load.
LO15	Design for dimension and strength of structural members subjected to various external loads, determine their deformation, and select the suitable material for a specific engineering application.
LO16	Analyze the stress and strain transformation at a point in two dimensions and determine the principal stresses/strains and their orientation.
LO17	Understanding the different method of energy method and its application on deflection solving
LO18	Apply different methods to solve for the deflection of statically indeterminate beam and the reactions, shear force and bending moment of statically indeterminate beams.
LO19	Analyzing a beam of different shape (curved) and a beam with symmetric and non – symmetrical cross sections
LO20	Apply different theories and technics to analyze the effect of torsion on Non-circular and thin-walled section.
LO21	Understand concepts, principles, laws, observations, and models of fluids at rest and in motion,
LO22	Understanding fluid behavior for engineering design and control of fluid systems,
LO23	Develop competence with mass, energy and momentum balances for determining resultant interactions of flows and engineered and natural systems,
LO24	Develop bases for correlating experimental data, designing procedures, and using scale models of fluid flows,
LO25	Knowing the nature of rotation, circulation, resistance (viscous, turbulent), boundary layers, and separation with applications to drag and lift on objects, and
LO26	Apply methods for computing head losses and flows in simple pipes and channels.
LO27	Solve steady and Transient, one-dimensional conduction heat transfer problems analytically
LO28	Use Numerical/finite difference approaches to numerically solve two dimensional, conduction and transient heat transfer problems
LO29	Understand fins and analyze heat transfer from extended surfaces
LO30	Understand the important of dimensionless parameters to convection heat transfer and basic principles underlying convection heat transfer
LO31	Understand thermal and viscous boundary layer heat transfer phenomena for laminar and turbulent flows
LO32	Analyzing heat transfer in various flow conditions (internal and external flows) by determining convection coefficient for a flow in/on a particular geometry.
LO33	Understand, select, size and analyze the performance of various types of heat exchangers
LO34	Understanding the basic concepts in refrigeration and air conditioning
LO35	Applying the working principles of vapor-compression, vapor-absorption, air and steam-jet refrigeration cycles.
LO36	Understanding the major equipment and the operations of vapor-compression, vapor-absorption, air and steam-jet refrigeration systems.
LO37	Applying the principles involved in the determination of cooling-load calculations and design of air-conditioning systems.
LO38	Understanding the air-Conditioning equipment and Air Distribution system
LO39	Understand the relationship between internal energy, heat and work as expressed by the First Law of Thermodynamics.
LO40	Apply the conservation of energy to thermodynamic systems
LO41	State and explain the Second Law of Thermodynamics
LO42	Explain how the Carnot cycle applies to heat engines and refrigeration cycles
LO43	Explain the concept of entropy.
LO44	Apply the concept of availability, irreversibility and the second law in defining the efficiency of a system
LO45	Understand the basic principles involved in mixture of ideal gases and gas-vapor mixtures.
LO46	Understand the combustion analysis, Adiabatic flame Temperature, LHV and HHV of fuels.
LO47	Understand the thermodynamic concepts to describe the performance of the individual components of an engineering system, e.g. a power plant, and then relate that information to the overall performance of the entire system.
LO48	Understand the basic principle of refrigeration.
LO49	Understand instrumentation and measurement in mechanical engineering field specifically electrical transducers for measuring force, displacement, velocity, temperature...
LO50	Use temperature, velocity, flow, force, acceleration, current, voltage measuring instruments
LO51	Calibrate instruments

LO52	Conduct simple experiments
LO53	Modeling common systems and identifying their responses to various inputs
LO54	Sensing (using sensors), Conditioning (using op-amps, transistors, ...), and Acquiring (using ADC, filters, ...) signals
LO55	Actuating (moving, pressurizing, ...) common systems
LO56	Controlling electromechanical systems using PLC or simple passive circuits
LO57	Understanding IC Engines characteristics and selection.
LO58	Understand processes and performance w.r.t power, efficiency and emission by application of first principles in thermodynamics, chemistry, heat transfer, fluid flow, and mechanical dynamics.
LO59	Describe and explain engine heat transfer and its relation to thermal loading of engine components and cooling.
LO60	Describe and explain engine friction, wear and lubrication
LO61	Sufficient knowledge on Engine management systems,
LO62	Describe methods for reduction of exhaust emissions, and their relations to fuel quality and engine performance
LO63	Sufficient knowledge on engine Trouble-shooting
LO64	Working knowledge in engine design w.r.t geometric parameters, performance parameters & pollutant emissions; Mechanical Design
LO65	Understanding of basic terms and definitions in mechanical design.
LO66	Explain the stress-strain characteristics of different materials.
LO67	Understand stress concentration and failure causes and types.
LO68	Comprehend and compare the various failure theories that suit the failure conditions.
LO69	Clarify the basic parameters and designing procedures for permanent and non-permanent joints.
LO70	Understand the major principles behind the function of mechanical springs and design different types of springs.
LO71	Make dimensional and material decisions for pressure vessels.
LO72	Understanding typical characteristics and design of principal machine elements (Shafts, gears, belts, bearings, and friction elements.
LO73	Understand the classification of turbo machines,
LO74	Understand the principles of operation of turbo machines,
LO75	Understand the energy losses and efficiencies of turbomachines,
LO76	Understand the performance characteristics of turbomachines, and its applications to determine their operating points,
LO77	Identify the dimensionless parameters of turbomachines and their significances/applications,
LO78	Understand the design guidelines of the rotors and housings of turbomachines
LO79	Knowing the different types of linkage mechanisms used in mechanical design;
LO80	Understanding the kinematic and kinetic analysis and design of machinery;
LO81	Applying the computer method for kinematic and kinetic analysis of mechanisms;
LO82	Design and analysis of cams, universal joints, governors, gear trains, flywheels and gyroscopes; and
LO83	Understanding the balancing
LO84	Know the different kinds of materials handling equipment,
LO85	Know the procedures for selection of material handling equipment suitable for a specific purpose, and
LO86	Know the steps in the design of hoisting and conveying equipment.
LO87	Understanding the basic concept of casting process, design of cast, casting defect and their remedies.
LO88	Understanding the basic principles and mechanisms of shearing and metal-forming process of selected processes;
LO89	Understanding the material consumption, forces and work done on selected machines and die design;
LO90	Understanding the basic principles and mechanisms of (plastics, rubbers, polymer matrix composite, ceramic and glass forming) processing.
LO91	Know the basic principles of powder metallurgy,
LO92	Understand basic traditional machining processes; describe their principles and tool geometry; evaluate tools wear/life, enumerate cutting force and power on traditional machine tools and measures to achieve optimization.
LO93	Understand advanced (i.e. subtractive and additive) manufacturing methods
LO94	Applying Principles of assembly and joining process in assembly;
LO95	Applying principles and operation of arc, gas, resistance, and other welding and joining processes.
LO96	Understand theoretical and practical aspects of maintenance practice in industrial setup;
LO97	Understand basics of damages of typical components of machinery and thereby help the student realize the state of damage of machinery;
LO98	Realize the use of the concepts of reliability, maintainability and availability in maintenance technology which are helpful in the prediction of plant performance;
LO99	Understand the organization of a maintenance department, maintenance planning and decision making processes;
LO100	Develop practical skill by providing some practical work of maintenance.
LO101	Understand the basic concept industrial management and organization;
LO102	Understanding the basic principle of industrial plant design;
LO103	Understand the effective material management;
LO104	Understanding the basic principle of management and resource allocation

Formulas

$$\text{Percent share for Theme } i = \frac{\text{Total Credit hour of the Theme } i}{\text{Total Credit hour of the Program } i} \times 100$$

$$\text{Percent share for Course } i = \frac{\text{Credit hour for Course } i}{\text{Total Credit hour of the Theme } i} \times \text{Percent share for Theme } i$$

$$\text{Total test items for Theme } i = \frac{\text{Total Credit hour for Theme } i}{\text{Total Credit hour for Program } i} \times \text{Total of items for the program}$$

$$\text{Total of items for course } = \frac{\text{Credit hour for Course } i}{\text{Total Credit hour for Theme } i} \times \text{Total number of items for Theme } i$$

7. Learning Outcomes in Terms of Three domain

The main program outcomes in terms of the cognitive, affective and psychomotor domains:

- Complete working knowledge of the fundamental principles that support Mechanical Engineering.
- Design and conduct experiments, as well as analyze and interpret data.
- Design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability.
- Ability to function in a multidisciplinary team.
- Ability to identify, formulate and solve engineering problems.
- Ability to communicate effectively.
- The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context.
- Engage in life-long learning and knowledge of contemporary issues.
- An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice.

8. Test Blueprint Table

Table 3 Test Blueprint Table

Theme	Course	Cr.Hrs.	General objective/Competency	Learning Outcome Code	Learning outcomes							No. of items
					Cognitive						Affective	
					Remembering	Understanding	Application	Analysis	Evaluation	Creation/Synthesis	Psychomotor	
Theme 1: Engineering Mechanics and Materials	Engineering Materials (I&II)	6	Ø Working knowledge of engineering materials Ø Engineering material identification/ prescription while differentiating availability and suitability Ø to use fault diagnosis tools and non-destructive testing	LO1		1						12
				LO2		1						
				LO3		1						
				LO4			1					
				LO5		1						
				LO6		1						
				LO7		1						
				LO8		1						
				LO9		1						
				LO10		1						
				LO11		1						
				LO12		1						
				LO13		0						
	Strength of Materials (I&II)	6	Ø Knowledge of designing products, usage, and repairing of machines tools and machineries. Ø testing and inspection of products Ø to determine compliance/ stiffness of products with specifications Ø to identify, formulate, and solve engineering problems Ø to design and conduct experiments, as well as to analyze and interpret data	LO14				2				11
				LO15					2			
				LO16				2				
				LO17		2						
				LO18			1					
				LO19				1				
				LO20			1					
Theme 2: Engine	Fluid Mechanics	3	Ø Fundamentals of Engineering Sciences, phenomena, and relationships of solid mechanics and	LO21		1						6

The me 3;			thermo-fluids, including their limitations. Ø to design and conduct experiments, as well as to analyze and interpret data										
				LO22		1							
				LO23						1			
				LO24						1			
				LO25		1							
				LO26			1						
	Heat Transfer	3	Ø Advanced mathematical techniques of calculus, differential equations and numerical methods Ø Fundamentals of Engineering Sciences, phenomena, and relationships of solid mechanics and thermo-fluids, including their limitations. Ø Knowledge of designing products, usage, and repairing of machines tools, material handling equipment, process equipment, fluid machines, power generation systems, refrigeration, air conditioning, and steam generation systems. Ø to identify, formulate, and solve engineering problems	LO27			1						
				LO28			1						
				LO29		1							
				LO30		1							
				LO31		1							
				LO32				1					
				LO33		0							
	Refrigeration and Air Conditioning	3	Ø Knowledge of designing products, usage, and repairing of machines tools, fluid machines, refrigeration, air conditioning, and steam generation systems. Ø Knowledge of relevant standards, codes, and regulations. Ø inspection and commissioning of equipment	LO34		2							
				LO35			1						
				LO36		1							
				LO37			1						
				LO38		1							
	Engineering Thermodynamics (I&II)	6	Ø Knowledge of designing products, usage, and repairing of machines tools, process equipment, fluid machines, power generation systems, refrigeration, air conditioning, and steam generation systems.	LO39		2							
				LO40			1						
				LO41		1							
				LO42		1							
				LO43		1							
				LO44			1						
				LO45		1							
				LO46		1							
				LO47		1							
				LO48		1							
	Inst rum enta	3	Ø Exposure to electrical and electronic circuits and machines.	LO49		2							5

Theme 5: Manufacturing, Maintenance and Industrial Management	Mechanism of Machinery	3	Ø Knowledge of machine elements and their respective design procedures. Ø to operate relevant computer software for design/ analysis/ optimization. to identify, formulate, and solve engineering problems	LO77	1														
				LO78		1													
				LO79		1													
				LO80		1													
				LO81			1												
				LO82						1									
				LO83		1													
	Materials Handling and Equipment	3	Ø Knowledge of machine elements and their respective design procedures. Ø Knowledge of designing products, usage, and repairing of material handling equipment. to determine the tools and equipment needed to do a job.	LO84		2													
				LO85		2													
				LO86		2													
	Manufacturing Engineering (I&II)	6	Ø Knowledge of metal fabrication processes and assembly processes. Ø design, sequence, and schedule production process of product Ø to determine the tools and equipment needed to do a job Ø to interpret written directions, specifications, plans, and drawings Ø testing and inspection of products or processes, and evaluating quality or performance.	LO87		2													
				LO88		2													
				LO89		1													
				LO90		1													
				LO91		1													
				LO92		1													
				LO93		1													
				LO94			1												
				LO95			1												
	Maintenance of Machinery	3	Ø Knowledge of maintenance procedures of machinery Ø Knowledge of the industrial principles of maintenance management Ø testing and inspection of products or processes, and evaluating quality or performance. Ø inspection and commissioning of equipment Ø to plan, control equipment maintenance, and determine life cycle costs	LO96		2													
				LO97		1													
				LO98	1														
				LO99		1													
				LO100						1									
	Industrial Management and Engineering Economy	3	Ø Principles and practices of personnel management and supervision. Ø Principles of plant layout design Ø Basic concepts of technical management and accounting, including project management	LO101		2													
				LO102		2													
				LO103		1													
				LO104		1													

	Total	63																	
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9. Conclusion

To meet the graduation profile, competency and learning outcome in the profession of mechanical engineering, exit exam competency selection and identifying core course for undergraduate mechanical engineering was drafted for 2015 E.C graduates previously. In test blueprint preparation, core competency and course learning outcomes mapping are effectively organized and the weights of the courses for the test and number of test items were analyzed by considering the learning outcomes and credit hours for each identified course to ensure the concept of content validity. The exam was expected to contain sixteen courses in five themes. The exam will expect to be given to thirty-six higher education institutes.